Manufactured Gradients: State AI Exposure and US Business AI Adoption at DeepSeek-R1

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Abstract

US states more exposed to AI — measured by the Anthropic Economic Index (AEI) usage-per-capita index — did not detectably accelerate business AI adoption relative to less-exposed states at the release of DeepSeek-R1 on 20 January 2025. In Census Bureau Business Trends and Outlook Survey (BTOS) state panels (2,727 state-wave cells, 51 states, 71 biweekly waves, 2023–2026), the change in each state's adoption-trend slope at R1 rises by +1.635 pp/yr per AEI index unit (HC1 se 1.458) — reproducing the scout estimate to the third decimal — but is statistically null: exposure-permutation $p = 0.257$. The same statistic at non-event placebo cutoffs brackets the headline (+1.56 at $t = 2024.30$, +2.24 at 2024.50), and a placebo *outcome* — remote work, whose federal return-to-office executive order was signed the same day as R1 — matches it (+1.97), with the weighted placebo version itself nominally significant ($p = 0.045$): the design manufactures gradients. Every nominally significant estimate in the battery traces to a known mechanism. Long-window negative gradients (−4.31 at $w = 1.0$) vanish under old-wording-only data — a splice artifact, because the nationally calibrated 1.553 rewording ratio misprices high-adoption, high-AEI states (corr +0.676). The two-way fixed-effects level effect (+0.527 pp, permutation $p = 0.011$) is a pre-trend: the high-minus-low gap was already growing at +0.600 pp/yr before R1, and state-specific trends flip the estimate to −0.38. The scan's localization at the first post-R1 wave is a mechanical rediscovery of the constructed treatment column. There is no credible evidence that high-AEI-exposure states accelerated AI adoption differentially at DeepSeek-R1: the BTOS state Bartik-style design, excluded from an earlier screen for want of exposure weights, stays dead even with real ones.

1 Introduction

The release of the DeepSeek-R1 reasoning model on 20 January 2025 [7] is the natural candidate event for a break in US business AI adoption: a companion sector-level analysis [10] of the Census Bureau's Business Trends and Outlook Survey (BTOS) found a sharp acceleration in AI-exposed sectors at exactly the first post-R1 survey wave, but attribution to R1 failed a placebo battery. This paper asks the cross-sectional version of the same question: did *states* whose workforces use AI more intensively accelerate measured business AI adoption differentially at R1? The design — state-level BTOS adoption interacted with an external AI-exposure measure — was excluded from an earlier automated screen for want of exposure weights; here we revive it with real ones.

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Both sides of the design rest on new measurement. The outcome is the BTOS AI use rate, a biweekly probability-sample estimate of the share of US firms using AI, introduced and validated by [3]. The exposure is the Anthropic Economic Index (AEI), which maps Claude conversations to economic tasks [8]; its geographic report [1] publishes a usage-per-capita index by US state, which we take as the cross-state exposure measure. Interacting differential exposure with a common shock is the logic that shift-share (Bartik) designs formalize [6], and its known failure modes discipline the analysis: exposure here is measured *after* the outcome window, so every estimate below is a descriptive gradient, never a causal Bartik coefficient; pre-existing differential trends can masquerade as treatment effects [13]; and slope changes at a single date in smooth aggregate series are spuriously significant often enough that placebo distributions at non-event dates, in the spirit of [5], are the appropriate yardstick. The per-state slope-change contrast is a kink-style estimand in the sense of the difference-in-kinks design [2], estimated and falsified with the `natex` toolkit [12], whose scan-based DiD [9] supplies a localization check. The battery converts one reproduced headline gradient and three nominally significant side results into a clean null plus three identified artifacts.

2 Data

The outcome panel combines three BTOS sources. The state sheets of the AI core workbook (DRB approval CBDRB-FY25-0425) give biweekly estimates and standard errors of the share answering "Yes" to Question 7 ("In the last two weeks, did this business use Artificial Intelligence (AI) in producing goods or services?"), waves 202319–202520. A fresh census.gov download extends the panel under the December-2025 rewording ("…in any of its business activities?"), waves 202524–202614; new-wording values are divided by 1.553, the national new/old measurement-RDD ratio estimated at the rewording cutoff by a companion paper [11], with an additive −6.04 pp variant as a diagnostic. The archived remote-work Question 6 (share of businesses with paid employees working from home at least one workday), waves 202417–202514, supplies a placebo outcome (1,221 state-wave cells). The assembled panel has 2,727 state-wave cells across 51 states (50 plus DC) and 71 waves, $t = 2023.710$–2026.507 in fractional years of collection-period midpoints; 24.7% of the state-by-wave grid is disclosure-suppressed, concentrated in small states.

Exposure is the AEI `usage_per_capita_index` per state [1], averaged over the 2026-04-01..05-01 and 2026-05-01..06-01 data windows (verified: CA $= (1.56 + 1.62)/2 = 1.59$); it ranges from 3.52 (DC) to 0.245 (WV). Two features bind the interpretation. First, the exposure window (April–June 2026) lies entirely *after* the outcome window, so exposure is endogenous by construction and no estimate below is causal, whatever its p-value. Second, the index measures Claude usage specifically, proxying all-AI exposure with attenuation and tech-hub confounding both plausible.

3 Design and Methods

Exposure gradient at a known cutoff. For each state s, local OLS slopes of the adoption share in calendar time are fit on each side of $t_0 = 2025.055$ (DeepSeek-R1; first post wave 202503) within $\pm w$ years, and the slope change $\Delta\kappa_s$ is the post-minus-pre difference — the right-minus-left kink convention of [2], with the cross-state gradient replacing the group contrast. The headline statistic is the cross-state regression of $\Delta\kappa_s$ on AEI exposure, b in pp/yr per index unit. The spec of record is $w = 0.5$, unweighted, restricted to states with at least 6 usable waves per side ($n = 36$; DC never enters — it is suppressed below the wave minimum). Inference is primary by exposure permutation (AEI values reassigned across states, $B = 10{,}000$, `default_rng(0)`), with

HC1 standard errors reported alongside [4]; windows $w \in \{0.5, 0.75, 1.0, 1.4\}$ are examined, and an SE-weighted variant fits $1/\text{se}^2$ WLS slopes from the published State SE sheets and combines them by inverse-variance meta-analysis.

Falsification battery. (i) placebo cutoffs at the non-event dates 2024.30 and 2024.50, at o1 (2024.6967, a real secondary event), at 2025.30, and at 2025.555 (a window that straddles the rewording splice); (ii) the placebo outcome (remote work) at the R1 cutoff — noting that the federal return-to-office executive order was signed on 2025-01-20, the *same day* as R1, so remote-work breaks at that date are expected for non-AI reasons; (iii) leverage checks dropping DC and CA, and a Spearman rank gradient; (iv) splice diagnostics rerunning the long-window gradients under the multiplicative (/1.553) splice, the additive (-6.04 pp) splice, and old-wording-only data ($t < 2025.8$).

Panel estimators. A tercile two-way fixed-effects (TWFE) level DiD interacts the top AEI tercile (17 states, cut 1.002) with post-R1, with wave and state fixed effects, CR1 standard errors by state [4], a placebo-in-space permutation of the treated set ($B = 2{,}000$), and a state-specific linear-trends variant. The `natex` survey pipeline's scan-based DiD (SuDDDS) [9] searches break dates and windows on the same panel (seed 0, normal model, AR(1)-by-unit permutation null) as a localization and role-assignment check.

Honest-inference notes, stated as run. HC1/CR1 intervals on smooth biweekly aggregates are potentially oversized, so permutation p is the primary inference throughout. The `natex` refusals are reported, not patched over: the SuDDDS anticipation test refused (Holm p all NaN), and all three effect estimators (dd, synthetic control, GESS) refused with "only 0 usable placebos; ≥ 5 required" on this unbalanced panel — the manual TWFE and permutation estimates below are the effect numbers of record.

4 Results

The scout number reproduces exactly, and it is null. At the spec of record ($w = 0.5$, unweighted, 36 states), the R1 slope-change gradient is $b = +1.635$ pp/yr per AEI index unit (HC1 se 1.458; classical se 1.428, matching the scout's "se 1.43, t 1.15"; $t_{\text{HC1}} = 1.12$; correlation 0.193), with exposure-permutation $p = 0.257$ (Table 1, panel A; Figure 1a). The SE-weighted version is +1.877 (HC1 se 1.922, $p_{\text{perm}} = 0.360$); the Spearman rank gradient is $\rho = 0.098$ ($p = 0.568$); dropping DC and CA gives +1.54 ($p = 0.326$) unweighted and +0.53 ($p = 0.804$) weighted. DC — the highest-exposure unit at AEI 3.52 — never enters the gradient sample in the first place, so no single unit drives the (non-)result.

Placebos bracket the headline. The identical statistic at non-event cutoffs returns +1.56 ($p = 0.277$) at $t = 2024.30$ and +2.24 ($p = 0.162$) at $t = 2024.50$ — magnitudes that *bracket* the R1 estimate — and -0.23 ($p = 0.889$) at o1 (Table 1, panel B; Figure 1b). The remote-work placebo outcome at R1 is +1.97 ($p = 0.288$), the same magnitude as the headline; its SE-weighted version is +4.39 (HC1 se 1.895) with permutation $p = 0.045$ — a nominally significant *placebo.* A design that produces R1-sized gradients at dates where nothing happened, and a significant gradient for an outcome whose own policy shock (the federal return-to-office executive order, signed 2025-01-20) merely shares R1's date, manufactures gradients; the null at R1 carries no information beyond that.

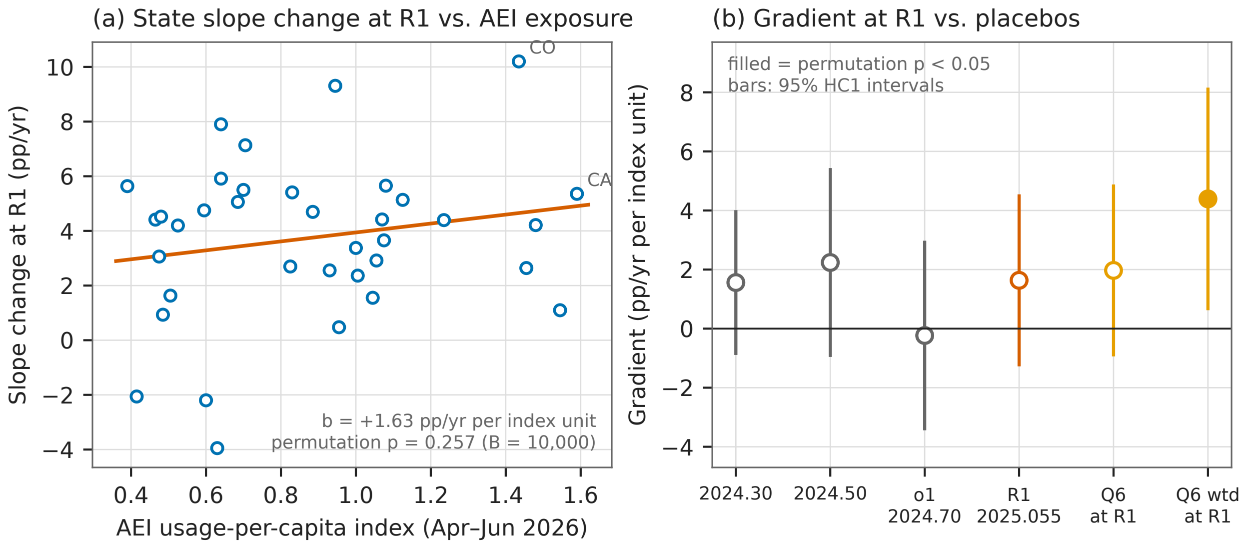

Figure 1: (a) State-level slope change in BTOS AI use at DeepSeek-R1 ($w = \pm 0.5$ yr, $n = 36$ states with ≥ 6 waves per side) against AEI usage-per-capita exposure, with the unweighted OLS fit: $b = +1.635$ pp/yr per index unit, exposure-permutation $p = 0.257$. (b) The same gradient statistic at non-event placebo cutoffs, at o1, at R1, and for the remote-work placebo outcome at R1, with 95% HC1 intervals; filled markers are nominally significant by permutation at the 5% level. The only filled marker in the battery is a *placebo* — the SE-weighted remote-work gradient at R1 (+4.39, $p = 0.045$).

Artifact 1: the long-window sign flip is the splice. Once the window reaches across the December-2025 rewording, the gradient turns significantly negative: -4.31 ($p = 0.011$) at $w = 1.0$ and -2.53 ($p = 0.011$) at $w = 1.4$ (Table 1, panel C). This is a measurement artifact, identified three ways. AEI exposure correlates +0.676 with the pre-R1 adoption level, so the nationally calibrated multiplicative ratio 1.553 misprices high-level (equivalently high-AEI) states; the additive -6.04 pp splice — wrong in the opposite direction — attenuates the same estimates to -2.92 and -1.07; and old-wording-only data collapse both to null (-0.87, $p = 0.279$ at $w = 1.0$; $+0.44$, $p = 0.513$ at $w = 1.4$). A placebo cutoff at 2025.555, placed so the window straddles the splice with no event nearby, yields -9.16 ($p = 0.00025$) multiplicative and still -5.81 ($p = 0.015$) additive: *neither* national correction is valid cross-state, so state-level gradients through the rewording — including any June-2026 extension of this design — are unidentified.

Artifact 2: the TWFE level effect is a pre-trend. The tercile TWFE DiD gives $\hat{\tau} = +0.527$ pp (CR1 se 0.210) with placebo-in-space permutation $p = 0.011$ (+0.547, $p = 0.038$ on the full panel) — nominally the strongest result in the study (Table 1, panel D). But the high-minus-low AEI gap was already growing at +0.600 pp/yr before R1 (HC1 se 0.180, 36 pre-R1 waves), which mechanically predicts roughly +0.3 pp of the estimate over the post window, and adding state-specific linear trends flips the $w = 0.5$ estimate to -0.38 (se 0.31). This is the textbook pre-trend failure [13]: the level DiD is descriptive only, and the trend-robust estimand — the slope-change gradient — is exactly the null of panel A.

Artifact 3: the scan localization is treatment rediscovery. The SuDDDS scan (seed 0, both panels) maximizes at LLR 154.5 for the empty subset — *all* states — at $t_0 = 2025.0877$,

Table 1: Exposure gradients, placebos, and panel estimates. Gradient rows: slope-change gradient b (pp/yr per AEI index unit) with HC1 se and exposure-permutation p ($B = 10{,}000$; splice diagnostics $B = 4{,}000$). TWFE rows: level effect (pp) with CR1 se by state and placebo-in-space permutation p ($B = 2{,}000$).

Specification	Estimate	se	p_{perm}	n
Panel A: AEI gradient at DeepSeek-R1 ($w = 0.5$ yr)				
unweighted (spec of record)	+1.635	1.458	0.257	36
SE-weighted (WLS + IV meta)	+1.877	1.922	0.360	36
unweighted, drop DC+CA	+1.544	1.651	0.326	35
SE-weighted, drop DC+CA	+0.531	2.316	0.804	35
Panel B: placebo cutoffs and placebo outcome ($w = 0.5$, unweighted)				
placebo cutoff $t_0 = 2024.30$	+1.560	1.224	0.277	33
placebo cutoff $t_0 = 2024.50$	+2.238	1.604	0.162	34
o1 cutoff $t_0 = 2024.6967$	−0.233	1.610	0.889	33
remote work (Q6) at R1	+1.970	1.458	0.288	51
remote work (Q6) at R1, SE-weighted	+4.392	1.895	0.045*	51
Panel C: long windows across the 2025-12 rewording (unweighted)				
$w = 1.0$, spliced (/1.553)	−4.305	1.676	0.011*	39
$w = 1.0$, additive splice (−6.04 pp)	−2.922	1.885	0.020*	39
$w = 1.0$, old wording only	−0.873	1.000	0.279	39
$w = 1.4$, spliced	−2.535	0.945	0.011*	40
$w = 1.4$, old wording only	+0.439	0.645	0.513	40
splice-window placebo $t_0 = 2025.555$, spliced	−9.161	2.150	0.00025*	39
splice-window placebo, additive	−5.806	2.639	0.015*	39
Panel D: tercile TWFE level DiD (pp)				
TWFE, $w = 0.5$	+0.527	0.210	0.011*	964
TWFE, full panel	+0.547	0.236	0.038*	2727
TWFE + state trends, $w = 0.5$	−0.382	0.315	—	964
pre-R1 high-minus-low gap slope (pp/yr)	+0.600	0.180	—	36 waves

Notes: * nominally significant at the 5% level by permutation. The $w = 1.4$ additive-splice gradient is −1.069 ($p = 0.095$). Panel D's TWFE p-values are placebo-in-space; the state-trends row and the pre-trend slope row report CR1 and HC1 standard errors respectively, with no permutation test run. Every nominally significant row is attributed to an identified artifact in the text: splice mispricing (panel C), a pre-existing differential trend (panel D), or a same-day policy shock to the placebo outcome (panel B).

the first post-R1 wave, with scan $p = 0.010$ under the AR(1)-by-unit null. Because the declared treatment column is the constructed `high_aei`×`post` indicator, a scan that lights up for all states at the post boundary is a mechanical rediscovery of that column, not outcome evidence — the same role-assignment failure documented in the companion sector study [10]. The composition check passes ($p = 0.707$); the anticipation test and all three effect estimators refused, as stated in Section 3.

5 Caveats and Conclusion

Four caveats bound the claims. First, *exposure is post-outcome*: the AEI index is measured April–June 2026, after every outcome wave used at the R1 cutoff, so even a robust gradient would have been descriptive, not Bartik-causal — the null is about this gradient design, not proof of zero differential adoption. Second, *exposure is Claude-specific*: as a proxy for all-AI exposure it is attenuated and plausibly confounded with tech-hub geography. Third, *coverage*: 24.7% of the state-wave grid is disclosure-suppressed, concentrated in small states; estimation samples run 33–40 states, and DC — the most exposed unit — is never estimable at the primary window. Fourth, *the calendar*: R1's release week was inauguration week, sharing its date with the federal return-to-office executive order (2025-01-20) and adjoining the Stargate announcement (2025-01-21), the AI-diffusion export rule (2025-01-13), and a BTOS sample-year rollover, so even a surviving gradient could not have been attributed to R1 alone.

The conclusion is a disciplined null. The headline gradient reproduces exactly and is statistically indistinguishable from the same statistic at dates where nothing happened; the one nominally significant gradient in the $w = 0.5$ battery belongs to a placebo outcome with its own same-day shock; and all three seemingly significant side results — the long-window negative gradients, the TWFE level effect, and the scan localization — dissolve under, respectively, old-wording-only data, state trends, and role-assignment scrutiny. These data provide no credible evidence that high-AEI-exposure states accelerated business AI adoption differentially at DeepSeek-R1, and the state-level splice problem implies that extending the design through the rewording cannot rescue it.

Reproducibility. All estimates were produced with `natex` v0.2.0 [12] at seed 0 from the frozen BTOS workbooks (published at `https://www.census.gov/hfp/btos`) and the public AEI state release [1]; no dataset files are committed. Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the headline gradient and the significant-placebo numbers against the numbers of record before drawing.

References

[1] Appel, R., McCrory, P., Tamkin, A., McCain, M., Neylon, T., and Stern, M. (2025). Anthropic Economic Index report: Uneven geographic and enterprise AI adoption. arXiv:2511.15080.

[2] Böckerman, P., Jysmä, S., and Kanninen, O. (2025). *Difference-in-Kinks Design*. IZA Discussion Paper No. 18313. `https://docs.iza.org/dp18313.pdf`

[3] Bonney, K., Breaux, C., Buffington, C., Dinlersoz, E., Foster, L., Goldschlag, N., Haltiwanger, J., Kroff, Z., and Savage, K. (2024). Tracking firm use of AI in real time: A snapshot from the Business Trends and Outlook Survey. NBER Working Paper No. 32319.

[4] Cameron, A. C., and Miller, D. L. (2015). A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, 50(2), 317–372.

[5] Ganong, P., and Jäger, S. (2018). A permutation test for the regression kink design. *Journal of the American Statistical Association*, 113(522), 494–504.

[6] Goldsmith-Pinkham, P., Sorkin, I., and Swift, H. (2020). Bartik instruments: What, when, why, and how. *American Economic Review*, 110(8), 2586–2624.

[7] Guo, D., Yang, D., Zhang, H., et al. (2025). DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature*, 645(8081), 633–638.

[8] Handa, K., Tamkin, A., McCain, M., et al. (2025). Which economic tasks are performed with AI? Evidence from millions of Claude conversations. arXiv:2503.04761.

[9] Herlands, W., McFowland III, E., Wilson, A. G., and Neill, D. B. (2018). Automated local regression discontinuity design discovery. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '18)*.

[10] Hillebrandt, H. (2026). A sharp bend, no verdict: Difference-in-kinks tests of US business AI adoption at DeepSeek-R1. Companion paper, natex paper collection. `https://haukehillebrandt.github.io/natex/btos-sector-did-r1/`

[11] Hillebrandt, H. (2026). The question is the treatment: A known-cutoff regression discontinuity at the BTOS AI-question rewording. Companion paper, natex paper collection. `https://haukehillebrandt.github.io/natex/btos-rewording-rdd/`

[12] Hillebrandt, H. (2026). *natex: automated natural-experiment discovery and estimation* (version 0.2.0). Software. `https://github.com/HaukeHillebrandt/natex`

[13] Roth, J. (2022). Pretest with caution: Event-study estimates after testing for parallel trends. *American Economic Review: Insights*, 4(3), 305–322.